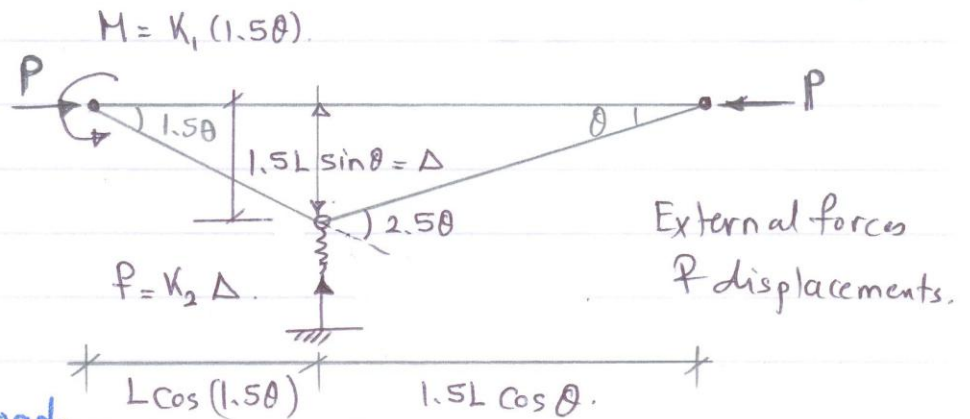


Model answer of Pind 2012

①

Question no. (1):-



1. Elastic buckling

Using Energy approach.

$$W = -P(1.5L + L - 1.5L \cos \theta - L \cos(1.5\theta))$$

$$= -P(2.5L - 1.5L \cos \theta - L \cos 1.5\theta)$$

$$U = \frac{1}{2} K_1 (1.5\theta)^2 + \frac{1}{2} K_1 (2.5\theta)^2 + \frac{1}{2} K_2 (1.5L \sin \theta)^2$$

$$\therefore \Pi = \frac{1}{2} K_1 \cdot 2.25 \theta^2 + \frac{1}{2} K_1 \cdot 6.25 \theta^2 + \frac{1}{2} K_2 \cdot 2.25 L^2 \sin^2 \theta$$

$$- P(2.5L - 1.5L \cos \theta - L \cos 1.5\theta).$$

$$\therefore \frac{d\Pi}{d\theta} = 2.25 K_1 \theta + 6.25 K_1 \theta + 2.25 K_2 L^2 \sin \theta \cdot \cos \theta$$

$$- PL \cdot (1.5) \sin \theta - PL(1.5) \cdot \sin 1.5\theta = 0.0$$

Small deflection

$$\sin \theta \approx \theta, \quad \sin 1.5\theta \approx 1.5\theta, \quad \cos \theta \approx 1$$

$$\therefore \frac{d\Pi}{d\theta} = 8.5 K_1 \theta + 2.25 K_2 L^2 \theta - (1.5 + 2.25) PL \theta = 0.0$$

for non trivial solution $\theta \neq 0.0$

$$\therefore 8.5 K_1 + 2.25 K_2 L^2 - 3.75 PL = 0.0$$

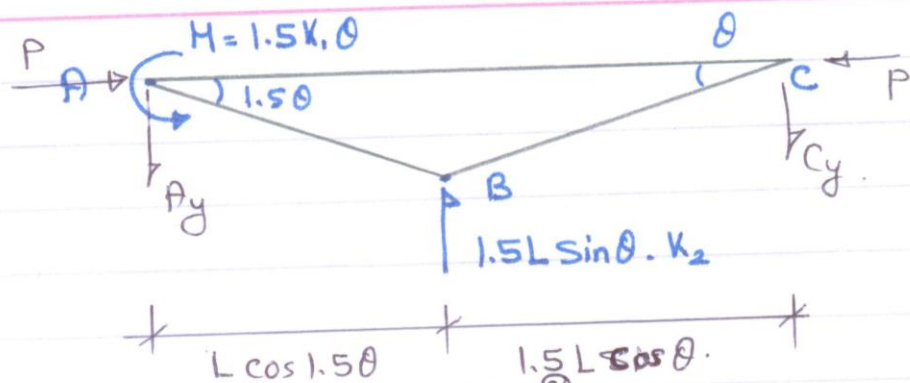
$$\therefore K_2 = K_1 / L^2$$

$$\therefore P_{cr} = \frac{10.75 K_1}{3.75 L} = \frac{2.9 K_1}{L} \quad \times$$

(2)

Bifurcation Approach:-

$$\sum M_C = 0.0$$



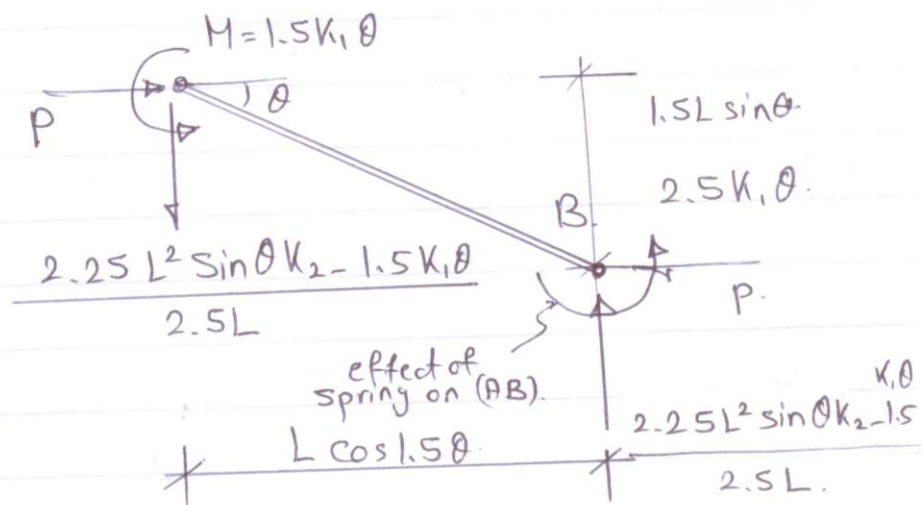
$$1.5L \sin \theta \cdot k_2 \cdot \underbrace{1.5L \cos \theta}_{(1)} - A_y (\underbrace{L \cos 1.5\theta + 1.5L \cos \theta}_{(2)}) - 1.5K_1\theta = 0.0$$

Since (θ) is very small

$$\therefore 1.5L \sin \theta \cdot k_2 \cdot \underbrace{(1.5L)}_{(1)} - \underbrace{2.5L}_{(2)} A_y - 1.5K_1\theta = 0.0$$

$$\therefore A_y = \frac{2.25L^2 \sin \theta k_2 - 1.5K_1\theta}{2.5L}$$

FBD of (AB):-



$$+\sum M_B = 0.0$$

$$1.5PL \sin \theta - 1.5K_1\theta - \frac{(2.25L^2 \sin \theta k_2 - 1.5K_1\theta)}{2.5L} L \cos 1.5\theta - 2.5K_1\theta = 0.0$$

θ is very small angle

$$\therefore 1.5PL\theta - 1.5K_1\theta - 2.5K_1\theta - \frac{(2.25 - 1.5)K_1\theta \times L}{2.5L} = 0.0$$

$$\therefore P_{cr} = \frac{4+0.30}{1.5} \frac{K_1}{L} = 2.90 \frac{K_1}{L}$$

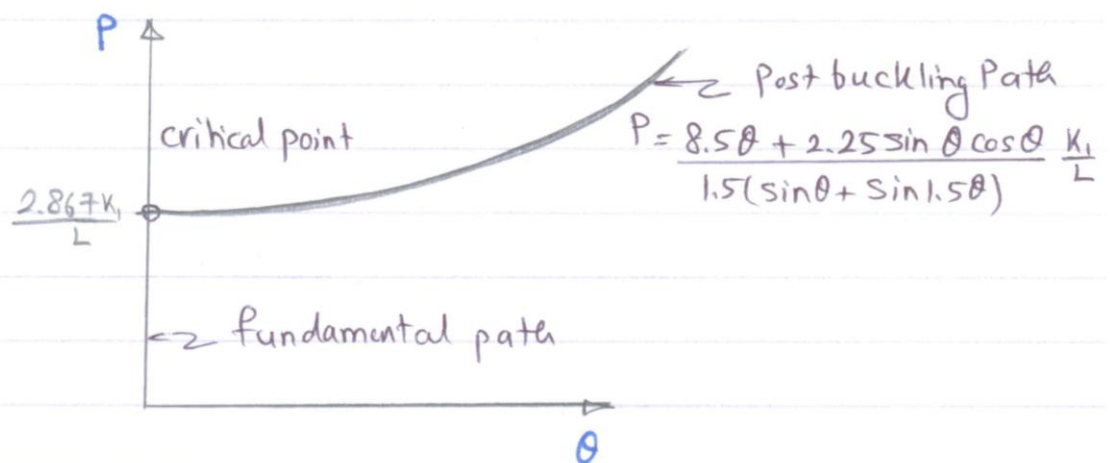
2. Load-deflection Curve :-

from energy approach for example

$$\frac{d\Pi}{d\theta} = 2.25 K_1 \theta + 6.25 K_1 \theta + 2.25 K_1 \sin \theta \cdot \cos \theta - 1.5 PL \sin \theta - 1.5 PL \sin(1.5\theta) = 0.0$$

$$\therefore P = \frac{8.5 K_1 \theta + 2.25 K_1 \sin \theta \cdot \cos \theta}{1.5 L (\sin \theta + \sin 1.5\theta)}$$

$$\therefore P = \frac{8.5 \theta + 2.25 \sin \theta \cdot \cos \theta}{1.5 (\sin \theta + \sin 1.5\theta)} \cdot \left(\frac{K_1}{L} \right)$$



3. State of equilibrium:-

$$\frac{d^2\Pi}{d\theta^2} = 8.5 K_1 + 2.25 K_1 (\cos^2 \theta - \sin^2 \theta) - 1.5 PL \cos \theta - 2.25 PL \cos(1.5\theta).$$

Fundamental path, $\theta = 0$

$$\therefore \frac{d^2\Pi}{d\theta^2} = 8.5 K_1 + 2.25 K_1 - 3.75 PL =$$

if $P > 2.87 \frac{K_1}{L}$ Unstable, $P < 2.87 \frac{K_1}{L}$ Stable

Along Post buckling Path:-

$$\frac{d^2\pi}{d\theta^2} = 8.5K_1 + 2.25K_1 [\cos^2\theta - \sin^2\theta] - \frac{(1.5K_1 \cos\theta + 2.25K_1 \cos 1.5\theta)(8.5K_1\theta + 2.25K_1 \sin\theta \cos\theta)}{1.5K_1 (\sin\theta + \sin 1.5\theta)}$$

$$\therefore \frac{d^2\pi}{d\theta^2} = K_1 \left[8.5 + 2.25 \cos 2\theta - \frac{(\cos\theta + 1.5 \cos 1.5\theta)}{\sin\theta + \sin 1.5\theta} \left[8.5\theta + \frac{2.25}{2} \sin 2\theta \right] \right]$$

for $0 \leq \theta \leq \pi/2$ $\frac{d^2\pi}{d\theta^2} = (+ve) \therefore$ (Stable)

At critical point:- $\theta = 0$ $P = \frac{2.87 K_1}{L}$

$$\therefore \frac{d^2\pi}{d\theta^2} = K_1 [8.50 + 2.25 - 1.5 \times 2.87 - 2.25 \times 2.87] = K_1 [0] = 0.0$$

$$\therefore \frac{d^3\pi}{d\theta^3} = -4.5K_1 \cos\theta \cdot \sin\theta - 4.5 \sin\theta \cos\theta K_1 + 1.5PL \sin\theta + 3.375 PL \sin(1.5\theta)$$

at $\theta = 0$ & $P = \frac{2.87 K_1}{L}$ $\frac{d^3\pi}{d\theta^3} = 0.0$

$$\therefore \frac{d^4\pi}{d\theta^4} = 4.5K_1 \sin^2\theta - 4.5K_1 \cos^2\theta + 4.5K_1 \sin^2\theta - 4.5K_1 \cos^2\theta + 1.5PL \cos\theta + 5.0625 PL \cos 1.5\theta$$

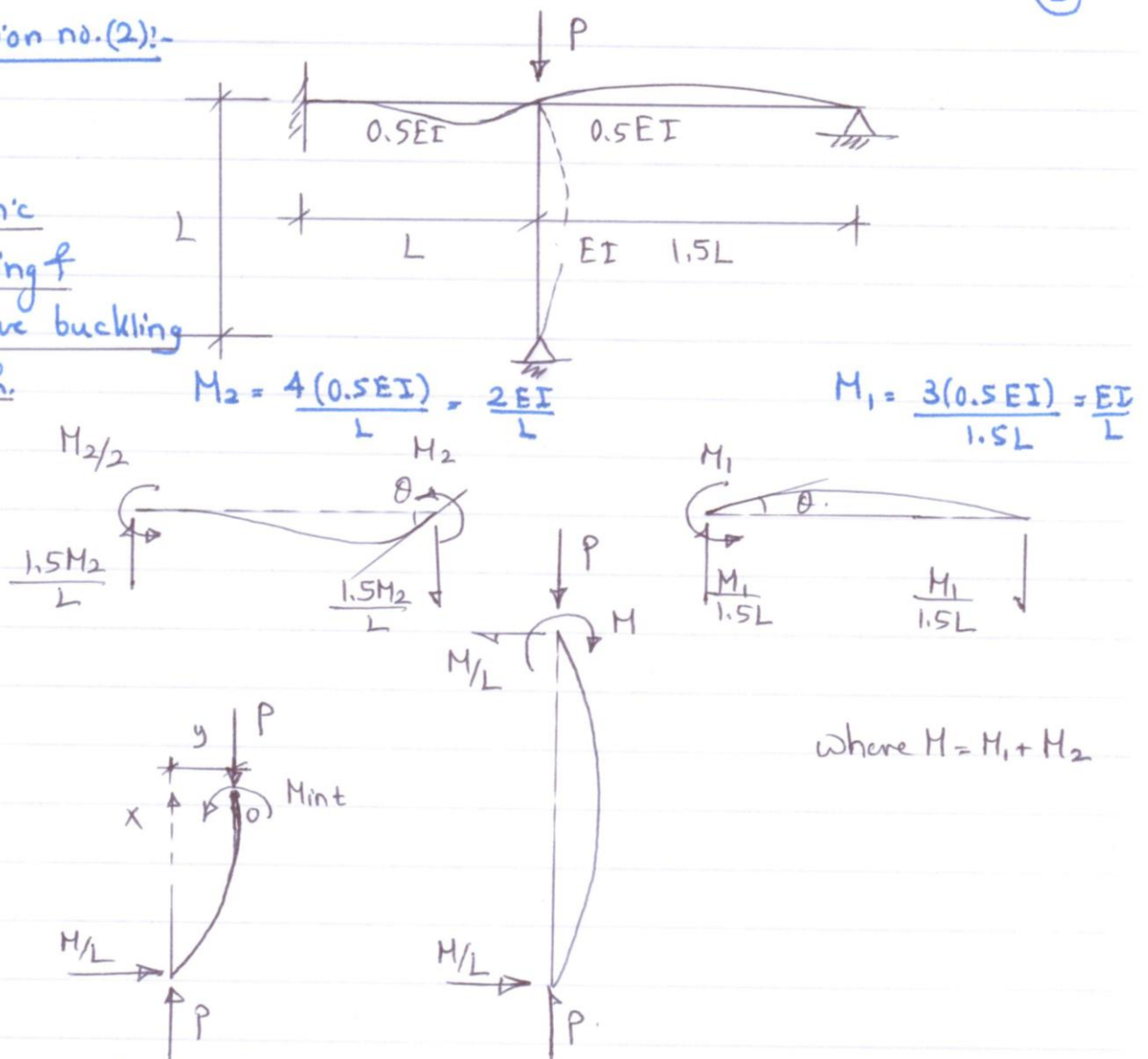
$$= 9K_1 \sin^2\theta - 9K_1 \cos^2\theta + 1.5PL \cos\theta + 5.0625 PL \cos 1.5\theta$$

$\therefore$ at $\theta = 0$ & $P = \frac{2.87 K_1}{L}$

$$\frac{d^4\pi}{d\theta^4} = 9.8125 K_1 > 0.0 \text{ (Stable).}$$

Question no. (2):-

Elastic
buckling &
effective buckling
length.



From equilibrium:-

$$\sum M_o = M_{int} + \frac{Mx}{L} - Py = 0.0 \quad M_{int} = -EIy''$$

$$-EIy'' - Py + \frac{Mx}{L} = 0.0$$

$$EIy'' + Py = \frac{Mx}{L} \quad \div EI \quad K^2 = P/EI$$

$$\therefore y'' + K^2y = \frac{Mx}{EIL}$$

$$\therefore y = A \sin Kx + B \cos Kx + \frac{Mx}{PL}$$

$$\text{at } x=0 \quad y=0 \quad \therefore B=0$$

$$\text{at } x=L \quad y=0 \quad \therefore 0 = A \sin KL + \frac{M}{P} \quad \therefore A = -\frac{M}{P \sin KL}$$

$$\therefore y = -\frac{M}{P} \frac{\sin Kx}{\sin KL} + \frac{Mx}{PL}$$

$$= \frac{M}{P} \left[\frac{x}{L} - \frac{\sin Kx}{\sin KL} \right]$$

$$y' = \frac{M}{P} \left[\frac{1}{L} - \frac{K \cos Kx}{\sin KL} \right]$$

$$\therefore y'(x=L) = \frac{M}{P} \left[\frac{1}{L} - \frac{K}{\tan KL} \right]$$

$$\therefore M_1 = \frac{EI\theta}{L}, \quad M_2 = \frac{2EI\theta}{L} \quad \therefore M = M_1 + M_2 = \frac{3EI}{L} \theta$$

$$\therefore \theta = \frac{ML}{3EI}$$

$$\therefore \theta = -y'(x=L)$$

$$\therefore \frac{ML}{3EI} = \frac{M}{P} \left[\frac{K}{\tan KL} - \frac{1}{L} \right]$$

$$\therefore \frac{K^2 L}{3} = \frac{KL - \tan KL}{L \tan KL}$$

$$\therefore (KL)^2 \tan KL - 3KL + 3 \tan KL = 0.0$$

$$\therefore KL = 3.7265$$

$$\therefore \sqrt{\frac{P}{EI}} \cdot L = 3.7265$$

$$\therefore \boxed{P = \frac{13.89 EI}{L^2}}$$

$$\therefore K = \sqrt{\frac{P_e}{P_{cr}}} = \sqrt{\frac{\pi^2 EI/L^2}{13.89 EI/L^2}} = 0.843$$

Using Alignment Charts:-

$G_c = 10$ (hinged)

$$G_B = \frac{(EI/L)}{\frac{0.5EI}{1.5L} \times 1.50 + \frac{0.5EI}{L} \times 2} = 0.67$$

end-hinged
end-fixed

From the alignment charts $K = \underline{0.85}$ (OK).

(7)

c. Capacity of column:-

ECP-LRFD $L_x = 0.85 \times 900 = 765 \text{ cm.}$

$$L_x / i_x = 765 / 16.5 = 46.36$$

$$\therefore \lambda_c = \frac{L_x / i_x}{\pi} \sqrt{\frac{f_y}{E}} = \frac{46.36}{\pi} \sqrt{\frac{2.4}{2100}} = 0.499 < 1.10$$

$$\begin{aligned} \therefore f_{cr} &= f_y (1 - 0.384 \lambda_c^2) \\ &= 2.40 (1 - 0.384 \times 0.499^2) \\ &= 2.1705 \text{ t/cm}^2 \end{aligned}$$

$$\begin{aligned} \therefore P_n &= A_g f_{cr} = 84\frac{1}{2} \times 2.1705 \\ &= \underline{183.41 \text{ ton.}} \end{aligned}$$

$$\therefore P_u = 0.80 P_n = \underline{146.73 \text{ ton.}}$$

Using Idealized Column with

$$(\sigma_r = 0.30 \sigma_y)$$

for Strong axis bending $\sigma_{cr} = \sigma_c \left[\frac{b_{fe}}{b_f} \right] = \sigma_c \sqrt{\frac{\sigma_y - \sigma_{cr}}{0.72}}$

$$\text{Where } \sigma_c = \frac{\pi^2 E}{(KL/r)^2} = \frac{\pi^2 \times 2100}{\left(\frac{0.85 \times 900}{16.5} \right)^2} = 9.642 \text{ t/cm}^2$$

$$\therefore \sigma_{cr} = 9.642 \sqrt{3.33 - 1.39 \sigma_{cr}}$$

$$\therefore \sigma_{cr}^2 = 92.97 (3.33 - 1.39 \sigma_{cr}).$$

$$\therefore \sigma_{cr}^2 + 129.23 \sigma_{cr} - 309.59 = 0.0$$

$$\therefore \sigma_{cr} = 2.353 \text{ t/cm}^2.$$

$$\therefore P_n = A_g \sigma_{cr} = 198.8 \text{ ton.}$$

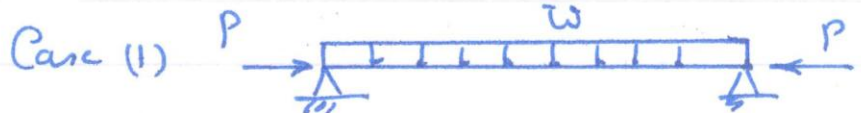
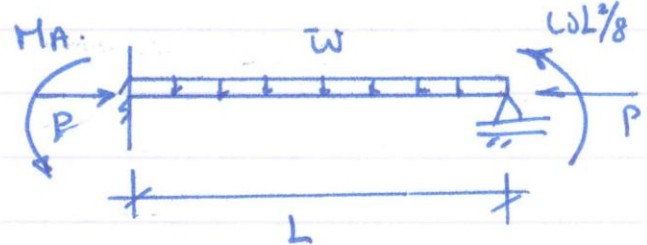
$$\therefore P_u = 0.80 P_n = \underline{159.05 \text{ ton.}}$$

Comment: The ECP-LRFD is more conservative as it was based on assuming $\sigma_r = 0.46\sigma_y$ and therefore provided less P_n compared to Idealized column with $\sigma_r = 0.30\sigma_y$ only.

Question no.(3):-

- location of Max. moment at support.

- The solution for $y(x)$ for case (1) is given and the solution of (-ve) case (2) is also given.



$$y_1 = \frac{w}{EI k^4} \left[\tan \frac{kL}{2} \sin kx + \cos kx - 1 \right] - \frac{w x}{2EI k^2} (L - x).$$

$$y_2 = - \left[\frac{-(M_A \cos kL + M_B)}{EI k^2 \sin kL} \sin kx + \frac{M_A}{EI k^2} \cos kx + \frac{M_A + M_B}{EI k^2} x - \frac{M_A}{EI k^2} \right].$$

Using the method of superposition (for constant P) we can obtain $y(x)$ in terms of M_A given that $(M_B = \frac{wL^2}{8})$.

$$\begin{aligned} \therefore y = & \left[\frac{w}{EI k^4} \tan \frac{kL}{2} + \frac{M_A \cos kL + wL^2/8}{EI k^2 \sin kL} \right] \sin kx \\ & + \left[\frac{w}{EI k^4} - \frac{M_A}{EI k^2} \right] \cos kx - \left[\frac{wL}{2EI k^2} + \frac{M_A + wL^2/8}{EI k^2} \right] x \\ & + \frac{w x^2}{2EI k} + \frac{M_A}{EI k^2} - \frac{w}{EI k^4}. \end{aligned}$$

(9)

Given that at (A) Fixed support $\therefore \frac{dy}{dx}(x=0) = 0.0$

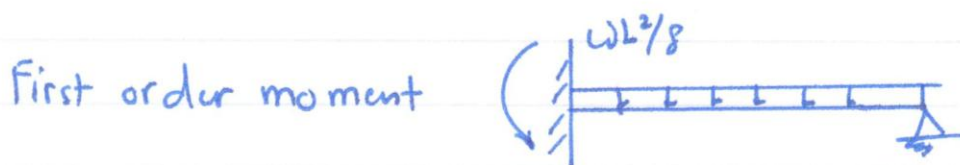
$$\therefore y' = \left[\frac{\omega}{EIk^3} \tan\left(\frac{kL}{2}\right) + \frac{MA \cos kL + \omega L^3/8}{EIk \sin kL} \right] \cos kx \\ - \left[\frac{\omega}{EIk^3} - \frac{MA}{EIk} \right] \sin kx - \left[\frac{\omega L}{2EIk^2} + \frac{MA + \omega L^3/8}{EIk^2} \right] \\ + \frac{\omega x}{EIk^2}$$

$$\therefore y'(0) = \frac{\omega}{EIk^3} \tan \frac{kL}{2} + \frac{MA \cos kL + \omega L^3/8}{EIk \sin kL} \\ - \frac{\omega L}{2EIk^2} - \frac{MA + \omega L^3/8}{EIk^2} = 0.0$$

$$\therefore MA \left[\frac{\cos kL}{\sin kL} - \frac{1}{kL} \right] = \frac{\omega L^3/8}{kL} + \frac{\omega L}{2k} - \frac{\omega L^3/8}{\sin kL} \\ - \frac{\omega \tan(kL/2)}{k^2}$$

$$\therefore MA \left[\frac{kL - \tan kL}{kL} \right] = \frac{\omega L^2}{8} \left[\frac{1}{kL} + \frac{4}{kL} - \frac{1}{\sin kL} \right. \\ \left. - \frac{8 \tan(kL/2)}{(kL)^2} \right]$$

$$\therefore MA = \frac{\omega L^2}{8} \cdot \frac{1}{kL} \left[5 - \frac{kL}{\sin kL} - \frac{8 \tan(kL/2)}{kL} \right] \cdot \frac{kL \tan kL}{(kL - \tan kL)}$$



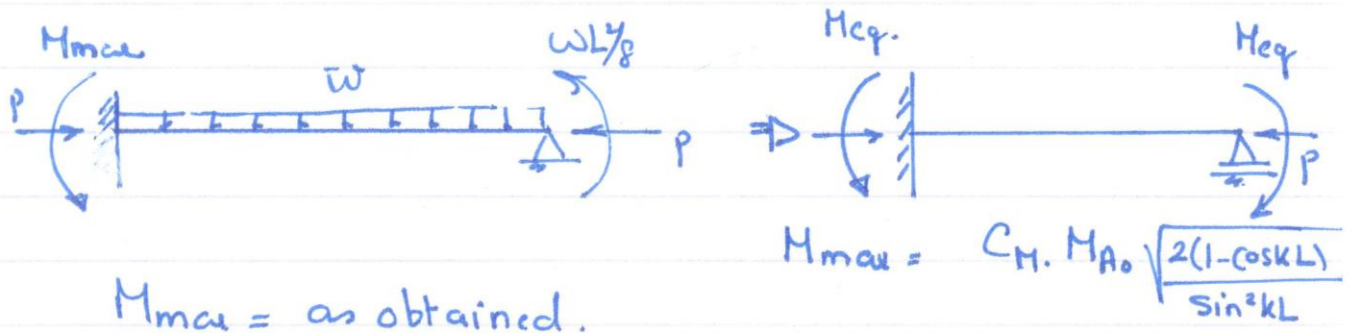
$$\therefore M_{A0} = \omega L^2/8 + \omega L^2/16 = \frac{3\omega L^2}{16}$$

(10)

$$\therefore M_A = \frac{3\omega L^2}{16} \times \frac{2}{3} \left[5 - \frac{KL}{\sin KL} - \frac{8 \tan(KL/2)}{(KL)} \right] \frac{1}{(KL - \tan KL)}$$

$$\therefore \text{Amp. factor (exact)} = \frac{2}{3} \left[5 - \frac{KL}{\sin KL} - \frac{8 \tan(KL/2)}{KL} \right] \frac{1}{(KL - \tan KL)}$$

Equivalent moment C_M in the form of $[1 + \psi (P/P_{ek})]$.



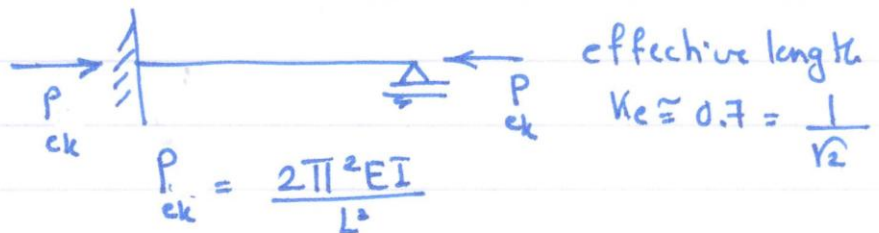
$$M_{\max} = C_M \cdot M_{A_0} \sqrt{\frac{2(1 - \cos KL)}{\sin^2 KL}}$$

$$\approx \frac{C_M \cdot M_{A_0}}{(1 - P/P_{ek})}$$

$$\approx \frac{M_{A_0} (1 + \psi P/P_{ek})}{(1 - P/P_{ek})}$$

$$\therefore \frac{3\omega L^2}{16} \cdot \frac{2}{3} \frac{\left[5 - KL/\sin KL - 8 \tan(KL/2)/KL \right]}{(KL - \tan KL)} = \frac{3\omega L^2}{16} \frac{(1 + \psi P/P_{ek})}{(1 - P/P_{ek})}$$

$$P = EI K^2$$

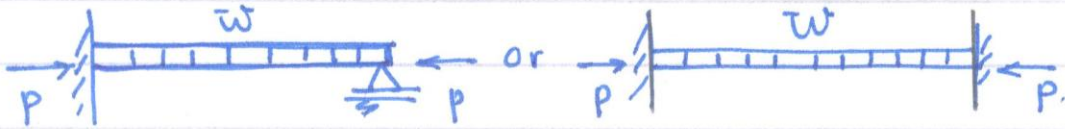


$$\therefore P/P_{ek} = \frac{EI K^2 L^2}{2\pi^2 EI} = \frac{(KL)^2}{2\pi^2}$$

$$\therefore \frac{2}{3} \frac{\left[5 - KL/\sin KL - 8 \tan(KL/2)/KL \right]}{(KL - \tan KL)} = \frac{1 + \psi (KL)^2/2\pi^2}{1 - (KL)^2/2\pi^2}$$

$\therefore$ Using Excel Sheet $KL = 0.05 \rightarrow 2\pi$ (fixed-fixed)
take average $(\psi) = \underline{\underline{-0.15}}$

- C_M recommended by ECP-LRFD
Table (2.11) page (2-24)



$$C_M = 1 - 0.40 (P/P_{ek})$$

i.e. $\psi = \underline{-0.40}$

- Moment amplification factor for $P/P_{ek} = 0.20$

$$\therefore P/P_{ek} = 0.20 \quad \therefore \frac{(KL)^2}{2\pi^2} = 0.20 \quad \therefore KL = 1.987$$

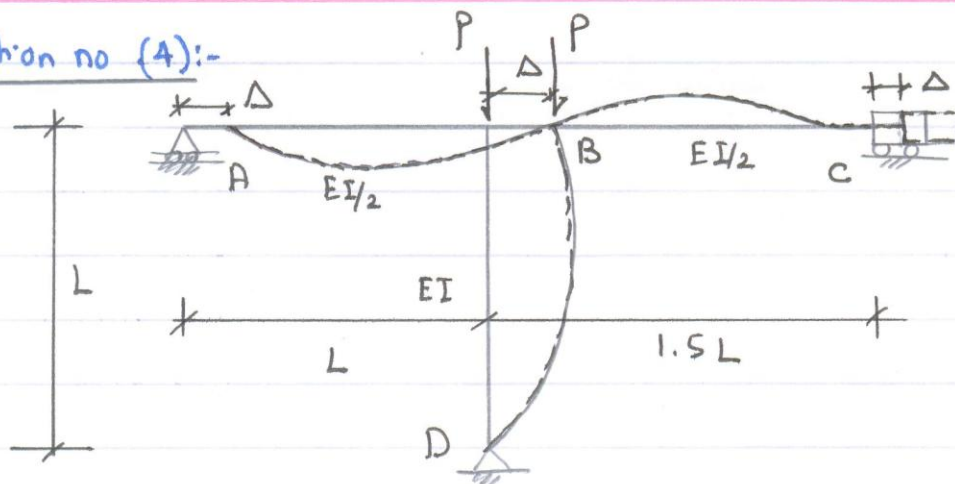
$$\therefore \text{Substitute for } M_{max}(\text{exact}) = \frac{3wL^2}{16} \times \underline{1.19}$$

$$\begin{aligned} \therefore \text{Substitute for } M_{max}(\text{design}) &= \frac{3wL^2}{16} \left[\frac{1 - 0.4 \times 0.20}{1 - 0.20} \right] \\ &= \frac{3wL^2}{16} \left[\underline{1.15} \right] \end{aligned}$$

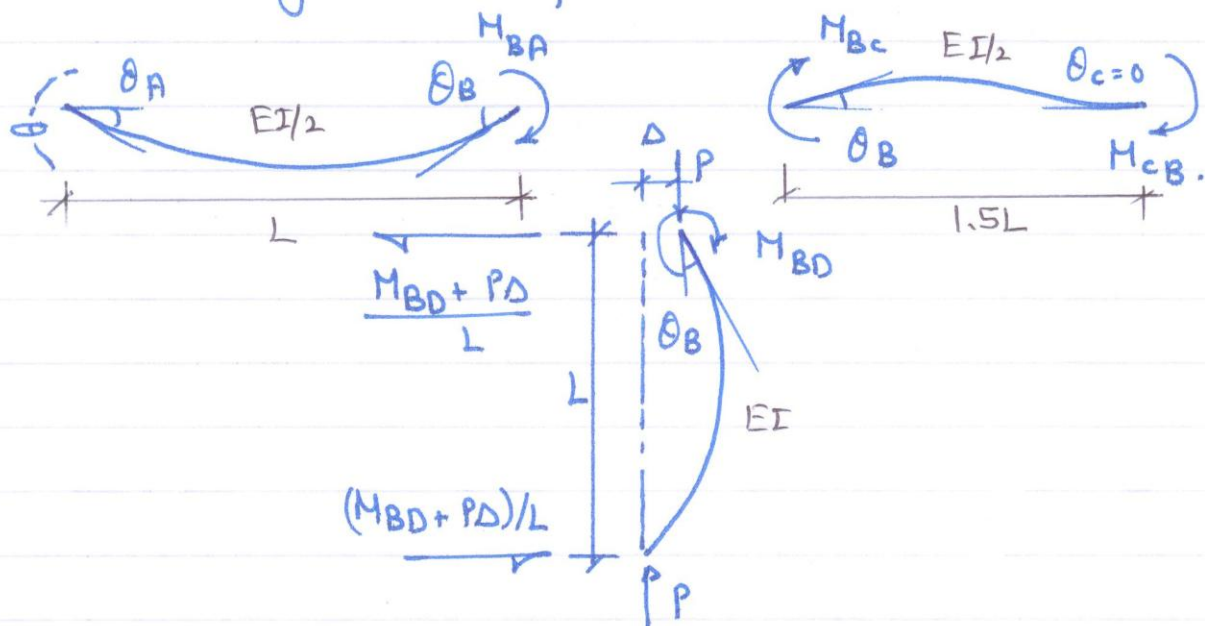
$$\begin{aligned} \text{Using } \psi = -0.15 \quad \therefore M_{max} &= \frac{3wL^2}{16} \left[\frac{1 - 0.15 \times 0.2}{1 - 0.20} \right] \\ &= \frac{3wL^2}{16} \left[\underline{1.21} \right] \end{aligned}$$

Comment:- Design equation underestimates M_{max} by 3%.
Whereas the proposed ψ provided conservative result (exceeding M_{max} by 1.6% only).

Question no (4):-



Elastic buckling load - Slope deflection method.



Beam (AB):- $S_{ii} = 4$ $S_{ij} = 2$ Neglect effect of axial force (No sway)

$$M_{AB} = \frac{(EI/2)}{L} [S_{ii} \theta_A + S_{ij} \theta_B] = 0 \quad \text{get } (\theta_A)$$

$$M_{BA} = \frac{(EI/2)}{L} [S_{ji} \theta_B + S_{jj} \theta_A]$$

$$\therefore M_{BA} = \frac{(EI/2)}{L} \left[S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right] \theta_B.$$

$$= \left(\frac{EI/2}{L} \right) \left[4 - \frac{(2)^2}{4} \right] \theta_B = \frac{1.50 EI \theta_B}{L}$$

Beam (BC) :- $S_{ii} = (4)$, $S_{ij} = 2$ (No axial force & No sway).

$$\therefore M_{Bc} = \frac{(EI/2)}{1.5L} [S_{ii} \theta_B + S_{ij} \theta_c] \because \theta_c = 0.0$$

$$= \frac{2EI}{1.5L} \theta_B.$$

Beam-Column. (BD) :- with sway. (Δ) , $M_{DB} = 0.0$

$$\therefore M_{BD} = \frac{EI}{L} \left[\left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right) \theta_B - \left[S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right] \frac{\Delta}{L} \right]$$

From equilibrium of moments:-

$$M_{BA} + M_{Bc} + M_{BD} = 0.0 \quad \text{--- (1)}$$

From equilibrium of hz. forces :-

$$(M_{BD} + P\Delta)/L = 0.0$$

$$\therefore M_{BD} + P\Delta = 0.0 \quad \text{--- (2)}$$

From (1)

$$\frac{EI}{1} \left[\left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right) \theta_B - \left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right) \frac{\Delta}{L} \right] + 2.83 \frac{EI}{L} \theta_B = 0.0$$

$$\therefore \left[\left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} + 2.83 \right) \theta_B - \left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right) \frac{\Delta}{L} \right] = 0.0$$

From (2)

$$\frac{EI}{L} \left[\left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right) \theta_B - \left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right) \frac{\Delta}{L} \right] + P\Delta = 0.0$$

$$\left[\left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} \right) \theta_B - \left(S_{ii} - \frac{(S_{ij})^2}{S_{ii}} - (KL)^2 \right) \frac{\Delta}{L} \right] = 0.0$$

$$\therefore \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \begin{Bmatrix} \theta_B \\ \Delta/L \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (14)$$

for non trivial solution, $\text{Det} = 0.0$

$$\therefore C_{11} \cdot C_{22} - C_{12} \cdot C_{21} = 0.0 \quad \text{Characteristic equation}$$

$$\text{Substitute } S_{ij} = \frac{KL \sin KL - (KL)^2 \cos KL}{2 - 2 \cos KL - KL \sin KL}$$

$$S_{ij} = \frac{(KL)^2 - KL \sin KL}{2 - 2 \cos KL - KL \sin KL}$$

by trial and error we get

$$\therefore KL = 1.177 = \sqrt{\frac{P}{EI}} \quad \therefore P_{cr} = \frac{1.385 EI}{L^2}$$

$$\therefore K_c = \sqrt{\frac{P_c}{P_{cr}}} = \sqrt{\frac{\pi^2 EI L^2}{L^2 \cdot 1.385 EI}} = \underline{\underline{2.67}}$$

Using alignment charts:-

$$\begin{array}{l} \text{Exact (theory)} \\ G_D = \infty \quad (\text{hinged}) \end{array}$$

$$\begin{array}{l} \text{Design (Code)} \\ G_D = 10 \quad (\text{hinged}) \end{array}$$

$$\begin{aligned} G_B &= \frac{EI/L}{\frac{0.5EI \times 0.5}{L} + \frac{0.5EI \times 0.67}{1.5L}} \\ &= 2.11 \end{aligned}$$

$$G_B = 2.11$$

$$\therefore K_c = \underline{\underline{2.10}}$$

$$\therefore K_c = \underline{\underline{2.67}}$$

Comment:- Code accounts for slight possible fixation provided by hinged support by taking ($G_D=10$). hence reduces (K_c) to (2.10)